

Apache Kafka Ultimate Master Guide

1. What is Kafka?

Apache Kafka is a distributed event streaming platform used for building real-time data pipelines and streaming applications.

2. Core Components

Producer: Sends messages. Consumer: Reads messages. Topic: Category for messages. Broker: Kafka server. Partition: Splits topic for scalability.

3. Producer Explained

A producer writes messages to Kafka topics. Messages can have a key (used for partitioning) and value (actual data).

4. Consumer Explained

Consumers subscribe to topics and read messages. They track offsets and can work in consumer groups for parallel processing.

5. Flow Overview

Producer -> Kafka Topic -> Consumer. Data is stored in partitions and read sequentially.

6. Key Concepts

Offset: Message pointer. Consumer Group: Load balancing. Retention: Data storage duration. Replication: Fault tolerance.

7. Real-world Example

E-commerce: Order service (Producer) sends orders, Kafka stores them, Shipping service (Consumer) processes deliveries.

8. Advantages

High throughput, fault tolerant, scalable, distributed, and real-time processing.

9. Simple Code Example (Concept)

Producer sends messages and Consumer polls them continuously from Kafka topic.